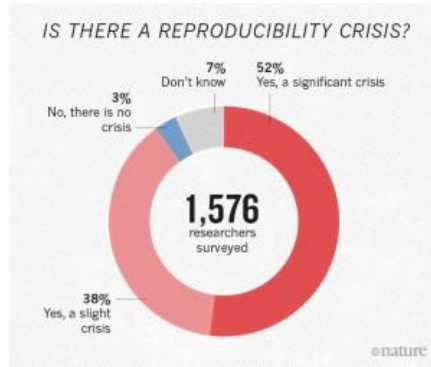


What is reproducibility?

- Key component of the scientific method, "cornerstone of science"
- Replicability: new data + same methods = same results
- **Reproducibility: same data + same analysis = same results**
- Repeatability: same researchers + same study + repeated experiments = same results
- "Reproducibility is more or less the ability to draw similar conclusions from replicates studies"
 - Diaba-Nuhoho, P., Amponsah-Offeh, M., *BMC Research Notes* (2021), <https://doi.org/10.1186/s13104-021-05875-3>

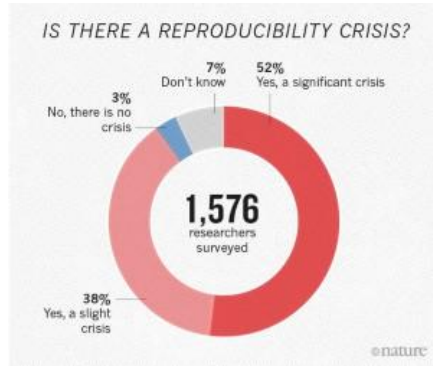
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 - ~1600 researchers: 52% significant crisis, 38% slight crisis (90% in total)

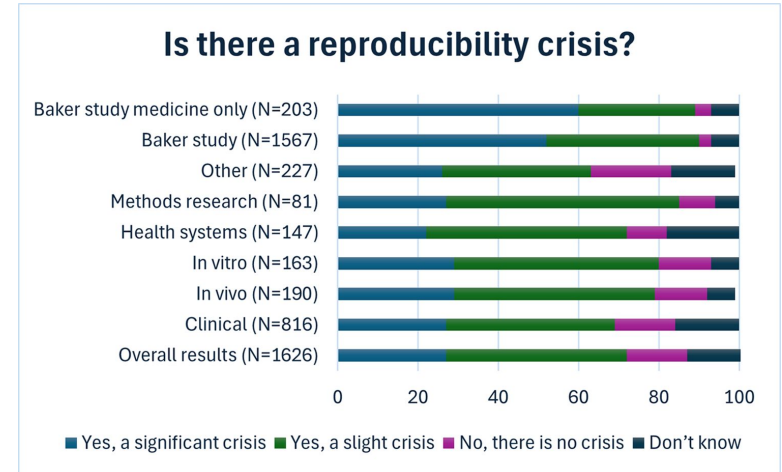


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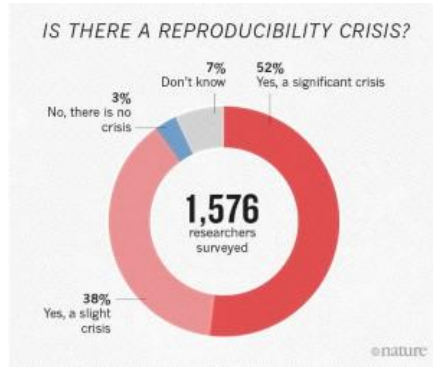


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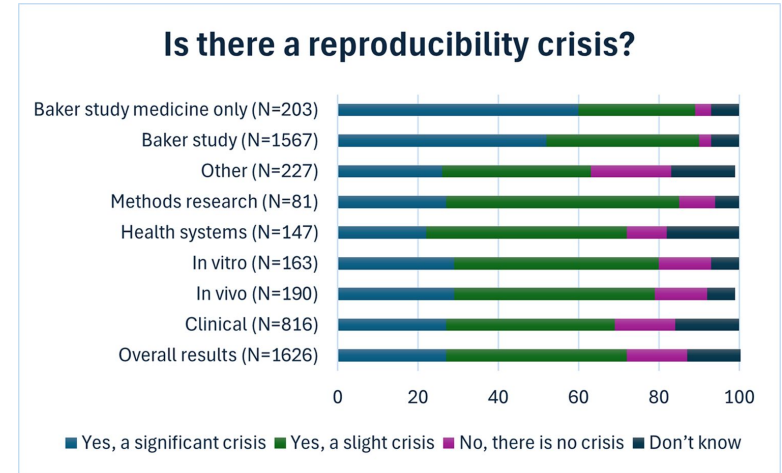


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- Han, H., *BMC Medical Genomics* (2025), <https://doi.org/10.1186/s12920-024-02072-6>
 - AI is revolutionizing biomedical data science at a fast pace but also brings a lot of questions about reproducibility

Why is that?

- Absence of knowledge/infrastructure
- Questionable research practices and fraud
- Statistical issues
 - Low statistical power
 - Statistical heterogeneity
- Publication system in science
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 - "Publish or perish"
 - Standards of reporting, open-access

COMPLETED

50 experiments

INITIATED

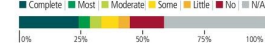
87 experiments

DESIGNED

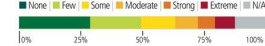
193 experiments

BARRIERS

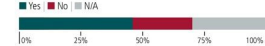
Modifications implemented



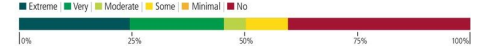
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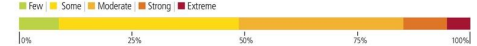
Reagents shared



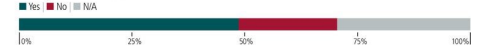
Authors helped



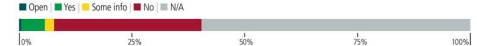
Protocol clarifications needed



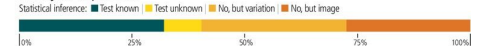
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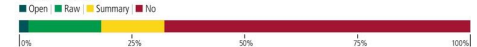
Code shared



Analysis reported



Data shared



Errington, T.M. *et al.*, *eLife* (2021),
<https://doi.org/10.7554/eLife.67995>

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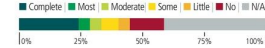
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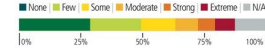
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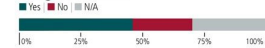
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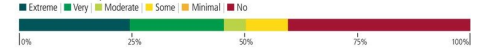
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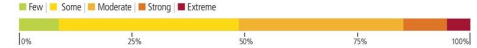
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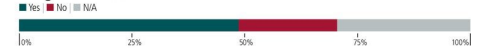
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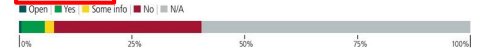
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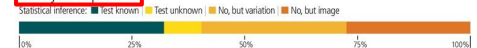
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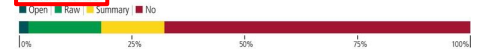
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What do Workflow Management Systems (WMS) bring?

- WMS can solve several hidden reproducibility issues:
 - Entirely:
 - OS
 - Version
 - Language
 - Readability
 - Availability
 - Partially (at least):
 - File format
 - Metadata
 - Parameters/options

What is the idea behind WMS?

- Provide reproducible and scalable infrastructures to set-up, perform, and monitor a defined sequence of tasks

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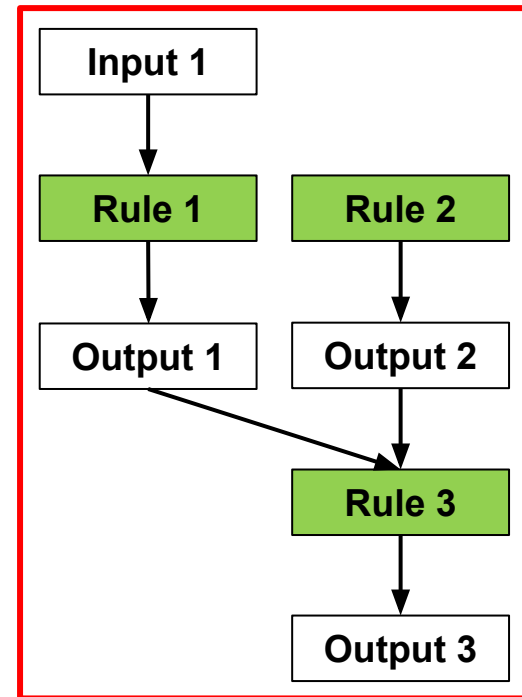
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- Easily deployable/executable **locally** or **remotely** (computation clusters and clouds)
- Integrated package management via **conda/mamba** (package manager) and **apptainer** (container manager)
- Once you have downloaded a workflow, it is easy to:
 - Run Snakemake in a **strictly controlled environment** (OS, software, versions, parameters...)
 - Efficiently and automatically reproduce analyses, results and figures

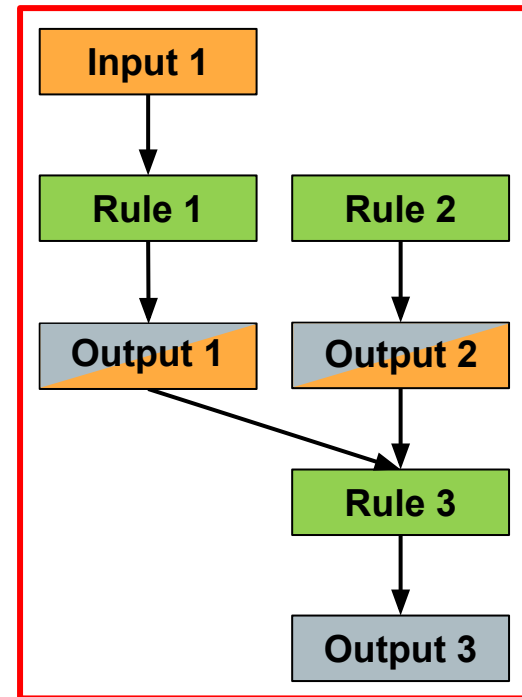
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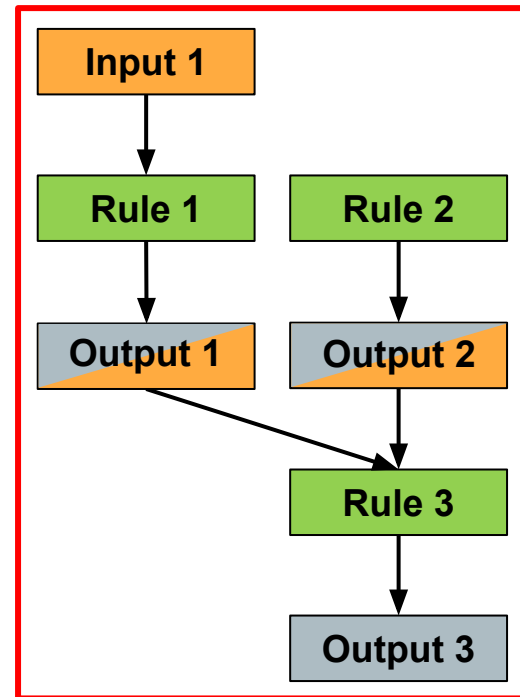
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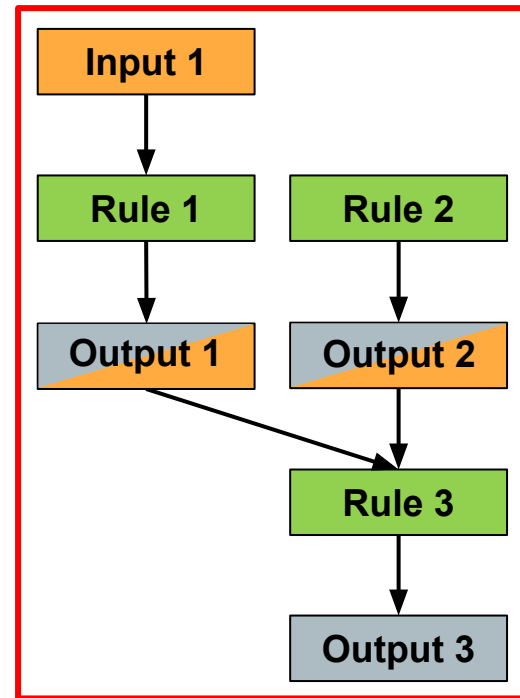
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- Directed Acyclic Graph (DAG) determined from the required outputs.



Why is a DAG useful?

- Skip parts of the DAG to avoid recomputing $\Rightarrow$ Save time and resources (CPU, memory, energy, money)
- Change/add inputs to existing analyses without re-running everything
- Resume running a workflow that failed part-way

What does Snakemake really look like?

```
rule rename_file:
    input:
        'data/file.txt'
    output:
        'results/renamed_file.txt'
    shell:
        'mv data/file.txt results/file_renamed.txt'
```

Snakemake lingo cheatsheet

- Snakemake keyword
- Rule name (user-defined)
- Snakemake directives
- Directives values:
 - Object
 - String (file path)
 - Instruction (command)
 - Numeric values (seen later)
- Wildcards
- Placeholders

```
rule rename_file:  
    input:  
        'data/{file}.txt'  
    output:  
        'results/{file}.txt'  
    shell:  
        'mv {input} {output}'
```

Exercises

- Throughout the day:
 - Develop a simple RNAseq analysis workflow, from reads (fastq files) to Differentially Expressed Genes (DEG)
- For this session:
 - Understand the structure of a Snakemake workflow
 - Create your first rules and Snakefile
 - Chain rules together
 - Run your first workflow

